

MODULE 4

Basic Principles, Design criteria & implementation issue of :

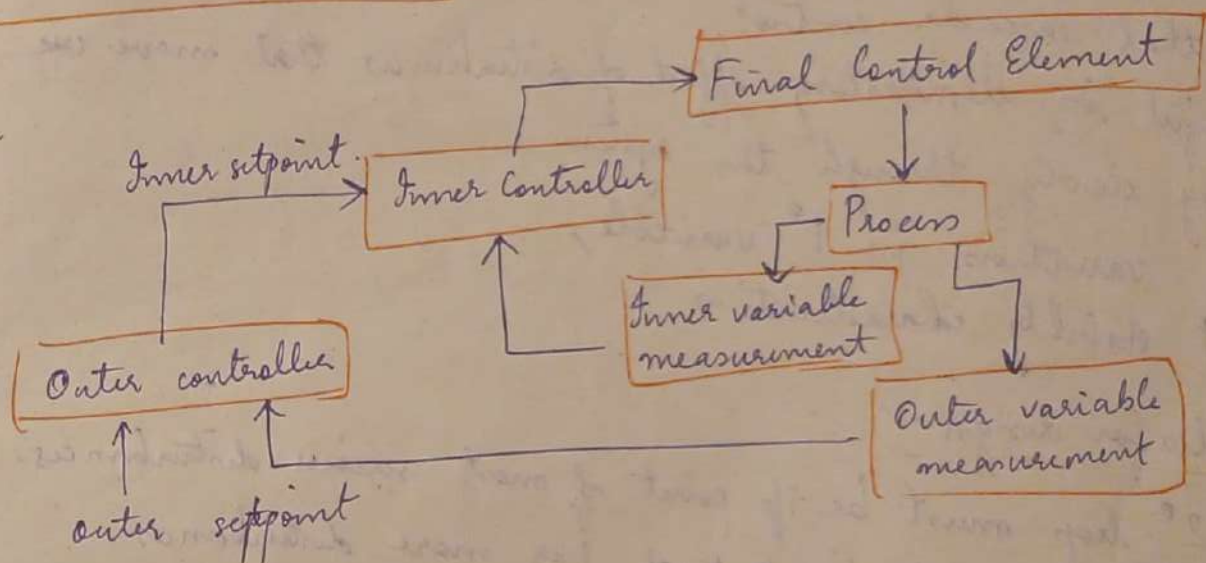
- ★ Cascade control
- ★ Feedforward control
- ★ feed back - feedforward control
- ★ Ratio control
- ★ Selective control
- ★ Split range control
- ★ Examples of any special features of the individual loop.
industry applns.
- ★ Non linear elements in loop:
 - Limiters
 - Dead zones
 - Backlash
 - Dead band velocity limiting
 - Negative resistance -

P & I DIAGRAMS.

↳ Piping & instrumentation

———— connection to process, instrument, supply.
// // pneumatic signals.
- - - - electrical signals.
- - - - differential

CASCADE CONTROL



- one measured variable
- one manipulated variable
- more than one measurement
- op of C_1 is sp of C_2

Basic principle: If 2^o variable responds quicker to disturbance than 1^o variable, then we can capture & nullify disturbance effect before it propagates into 1^o variable.

Inner loop is secondary

Outer loop is primary

- A disturbance causes change in op of primary controller (C_1) which causes change in set point of inner loop.
- Even tho there's no change in measured value of inner loop variable, inner loop (C_2) experiences error signal & produces new op by virtue of ΔSP .
- 2 feedback loops are nested ~~over~~ together.
- inner loop has effect of reducing time lag in outer loop, so that cascade control
- useful in eliminating effect of disturbances that move ~~very~~ very slowly through the system:
- $\downarrow$ variation in 1^o variable,
 $\uparrow$ stability characteristics.

Rules for design

- 1) 2^o loop must be ip point of most serious disturbances.
- 2) 2^{ad} loop must $\downarrow$ effect of 1 or more disturbances.
- 3) The 2^{ad} loop must be fast by introducing ~~minor~~ only minor lags of complete control system.
 $\hookrightarrow$ 2^o loop must be at least 3 times faster than 1^o loop.
- 4) 2^o variable must provide stable performance w/ narrow proportional band.

Advantages

- 1) Better control of 1^o variable.
- 2) 1^o var. is less affected by disturbances.
- 3) faster recovery from ~~disturbances~~ disturbances.
- 4) $\uparrow$ natural system frequency.
- 5) improves dynamic performance.

T : measured variable
 ↳ temp. of reactor

F_c : coolant flow in rate
 ↳ manipulated variable

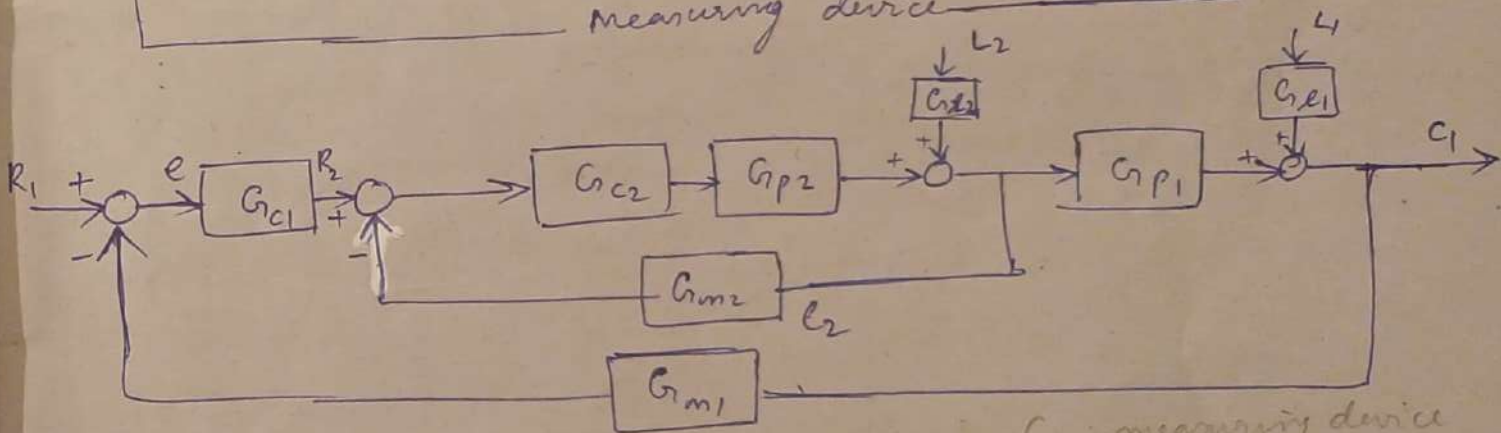
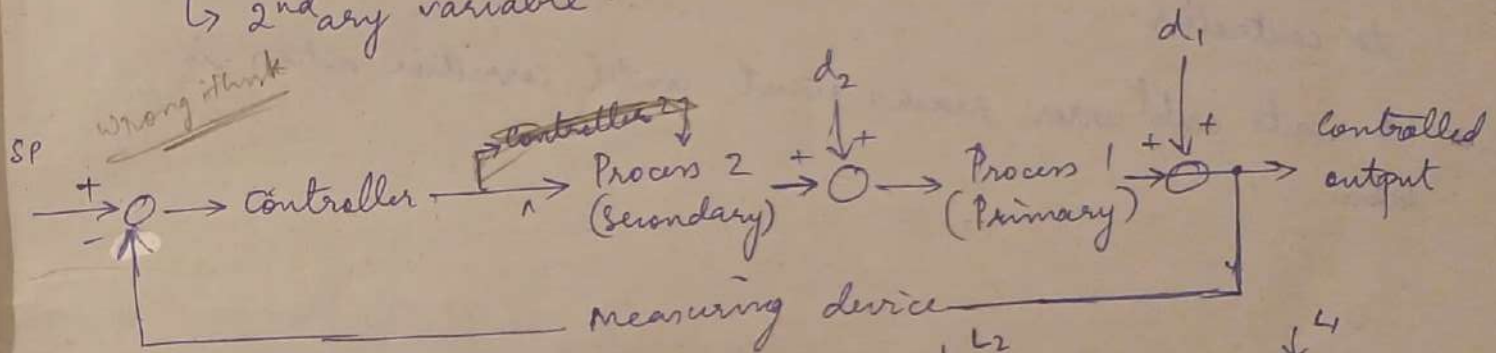
T_c : coolant temp.
 ↳ inner variable

Process 1: reaction in tank.

T is primary variable to be controlled

Process 2: jacket around CSTR and its output T_c

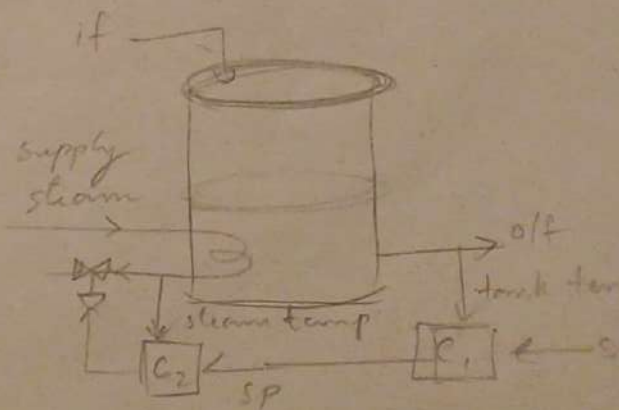
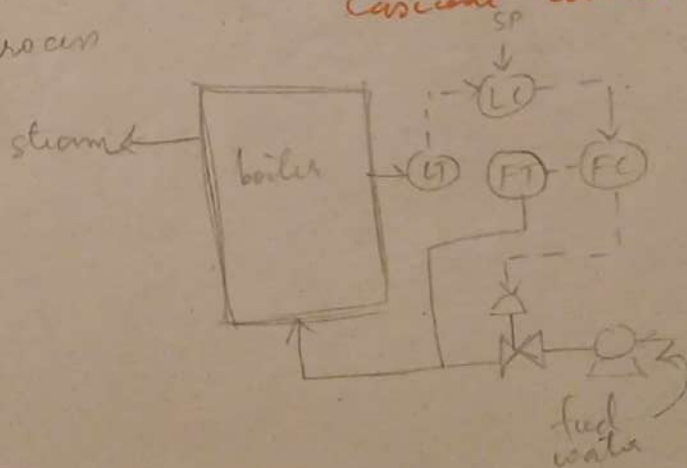
↳ 2ndary variable



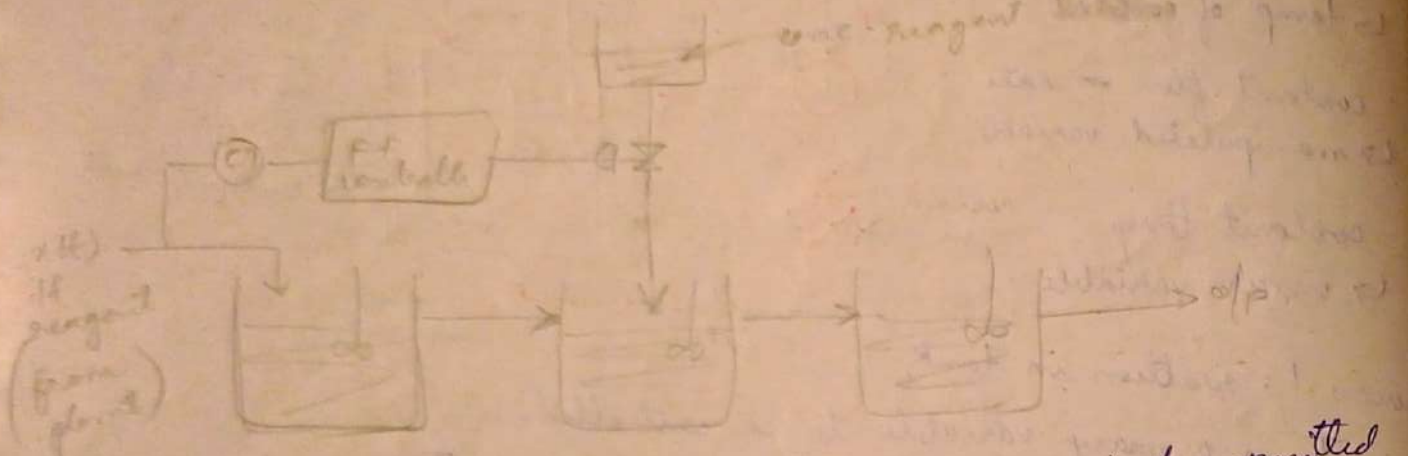
G_c : controller

G_p : process

Cascade control loop.

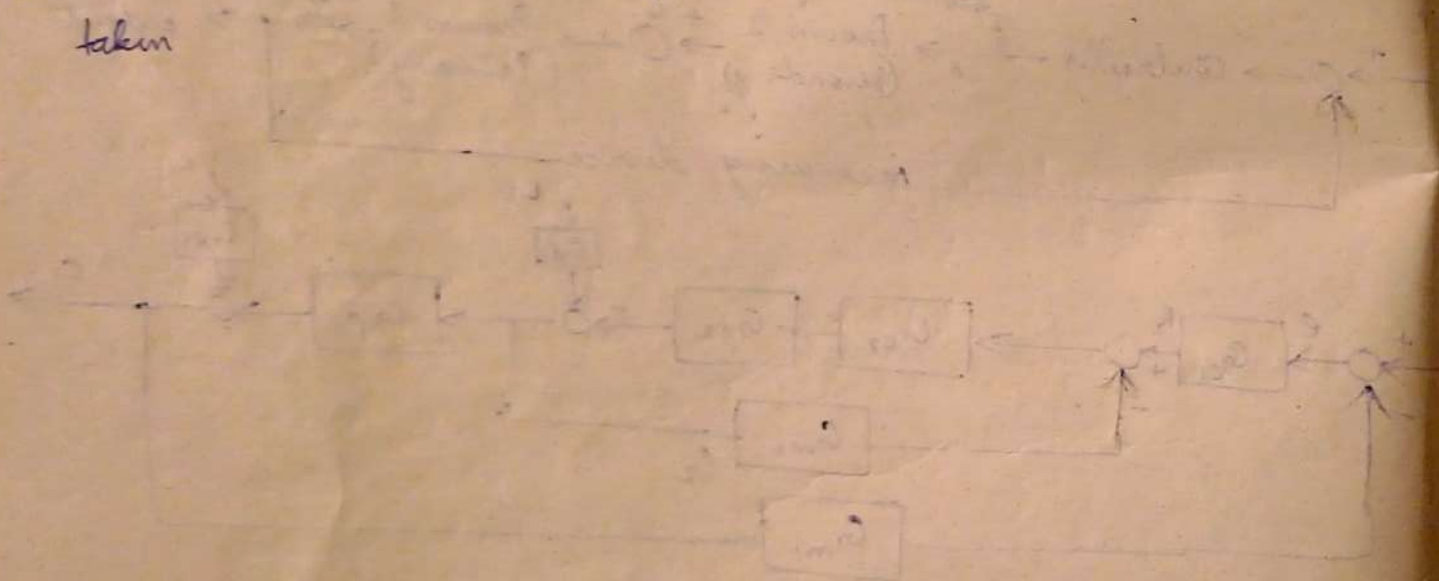


FEED FORWARD CONTROL

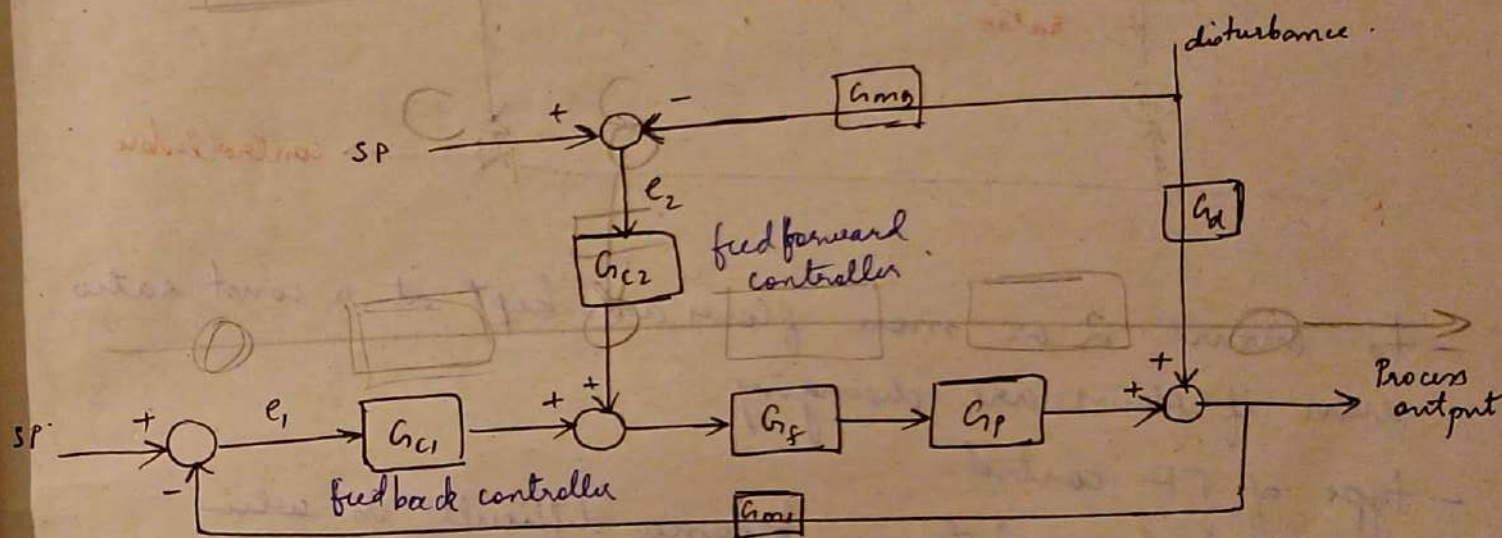


- δ/g which have potential to disturb sys. is transmitted to controller

* FB waits until error reaches plant until corrective action is taken

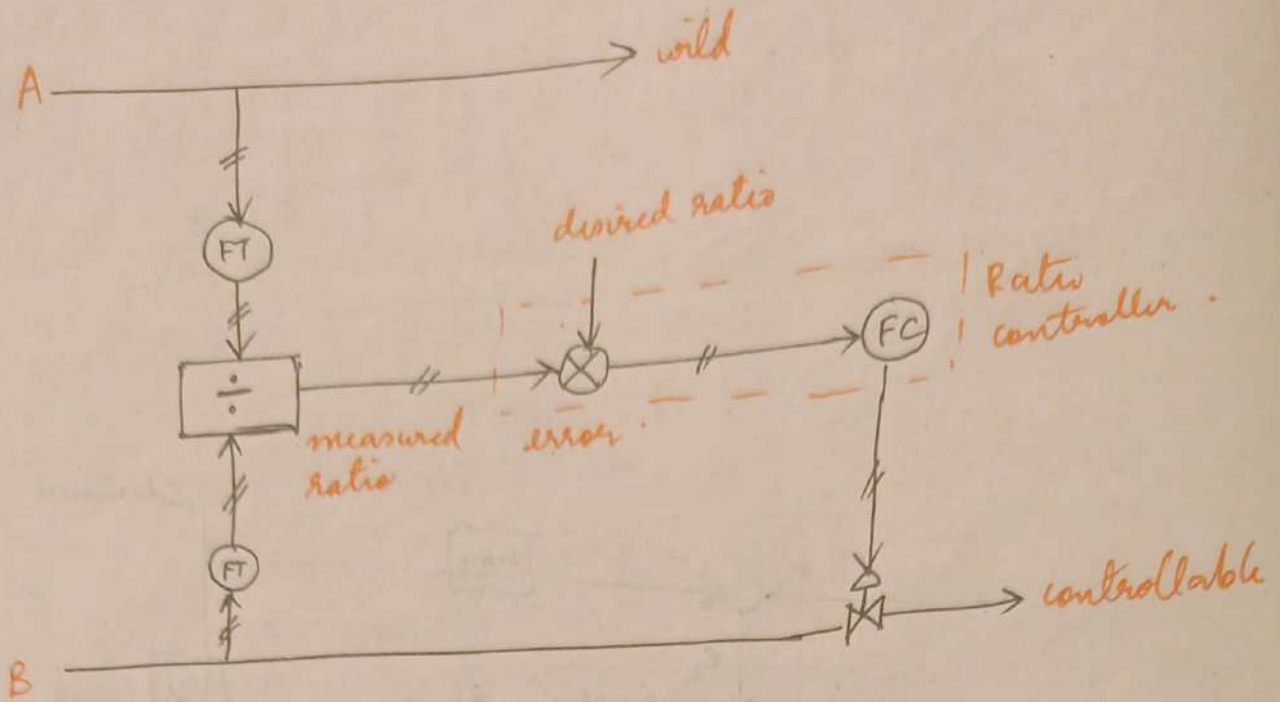


FEEDFORWARD - FEEDBACK CONTROL

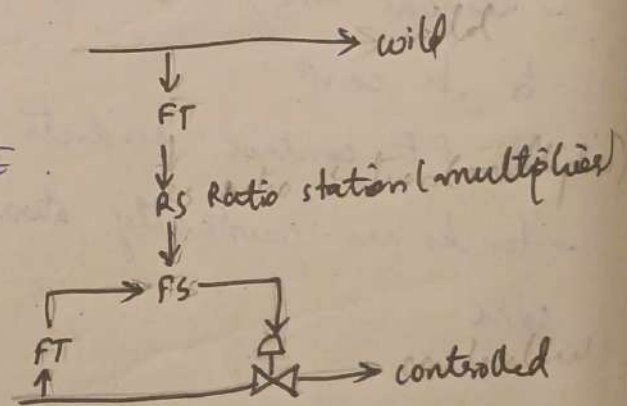
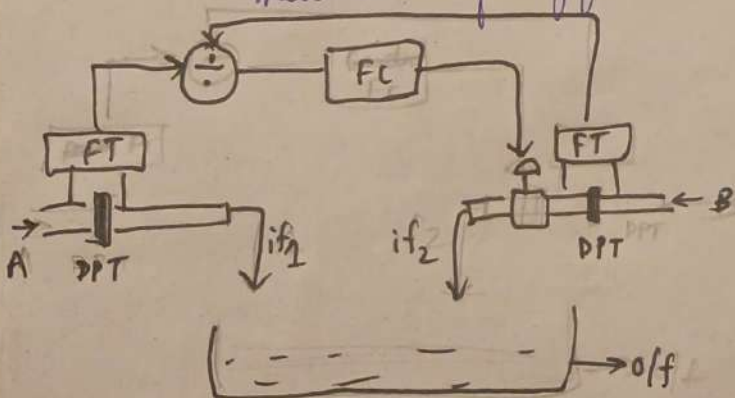


- anticipates change by detecting change in disturbance at input.
- reagent may have fluctuating conc. but o/p of plant needs to be const.
- so FF control predicts & works alongside FB control.
- tanks are constantly stirred so tank as a whole has uniform conc.

RATIO CONTROL



- to ensure 2 or more flows are kept at a const ratio even if flows are changing.
- type of FF control
- ~~on~~ control of 1 stream becomes difficult in ~~which~~ case, its flow is measured and then flow of other stream is controlled.
- egs: ratio of 2 reactants ^{entering} a reactor at a desired ~~ratio~~ ratio
- pharmaceutical drugs.
- air & fuel mixture in ICE.
- neutralise pH of fluids.

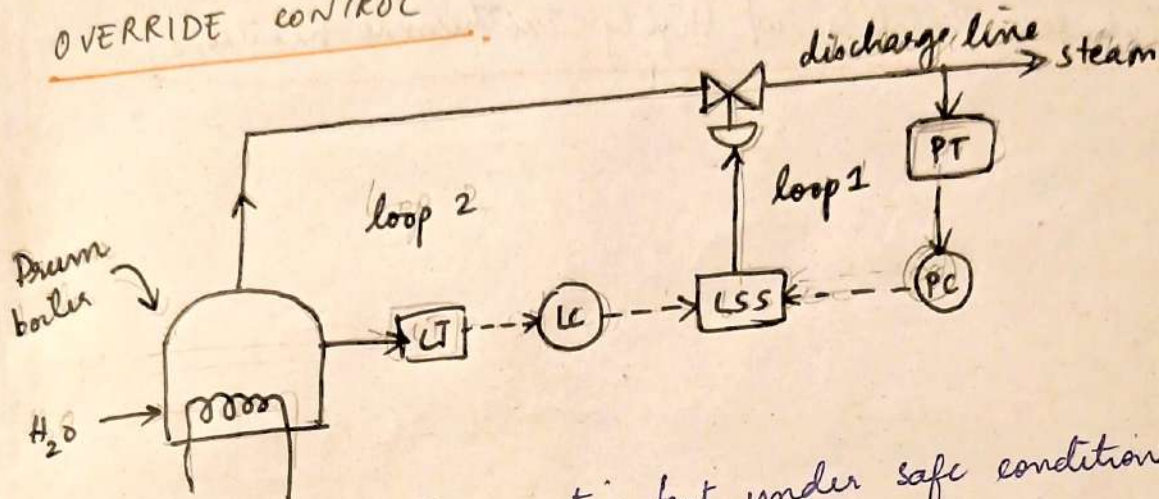


PPT: orifice plate or something.

SELECTIVE CONTROL

when manipulated variable $<$ controlled variable -
selective control is used, which transfers control action from
one controlled variable to another

OVERRIDE CONTROL



- maintains process in operation but under safe conditions.

HSS: high selector switch.

LSS: low selector switch.

$$e = SP - \text{measured}$$

here:

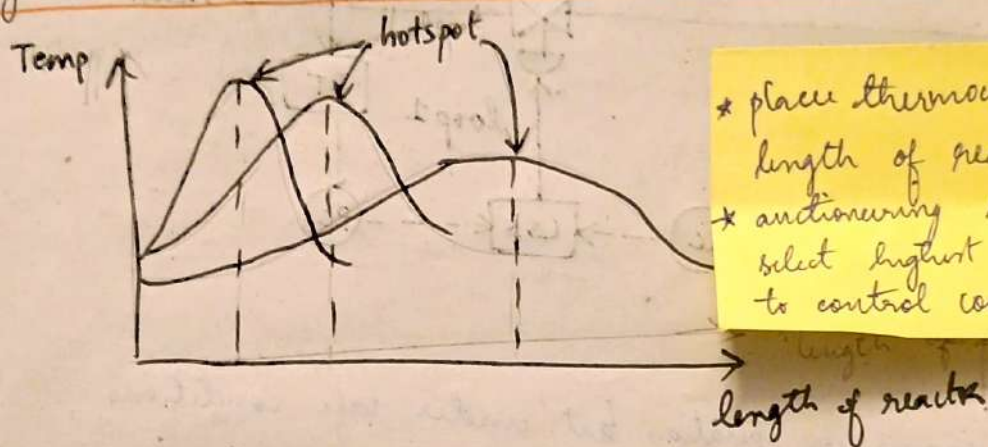
- protects drum boiler from overheating
- usually pressure in a boiler is controlled by use of a pressure loop control loop (loop 1)
- water in boiler must not be beyond a certain level to ensure heating coil is immersed in water & doesn't burn out.

- whenever level $\downarrow$ below certain limit, LSS switches control action from pressure control to low level control & closes valve on discharge line

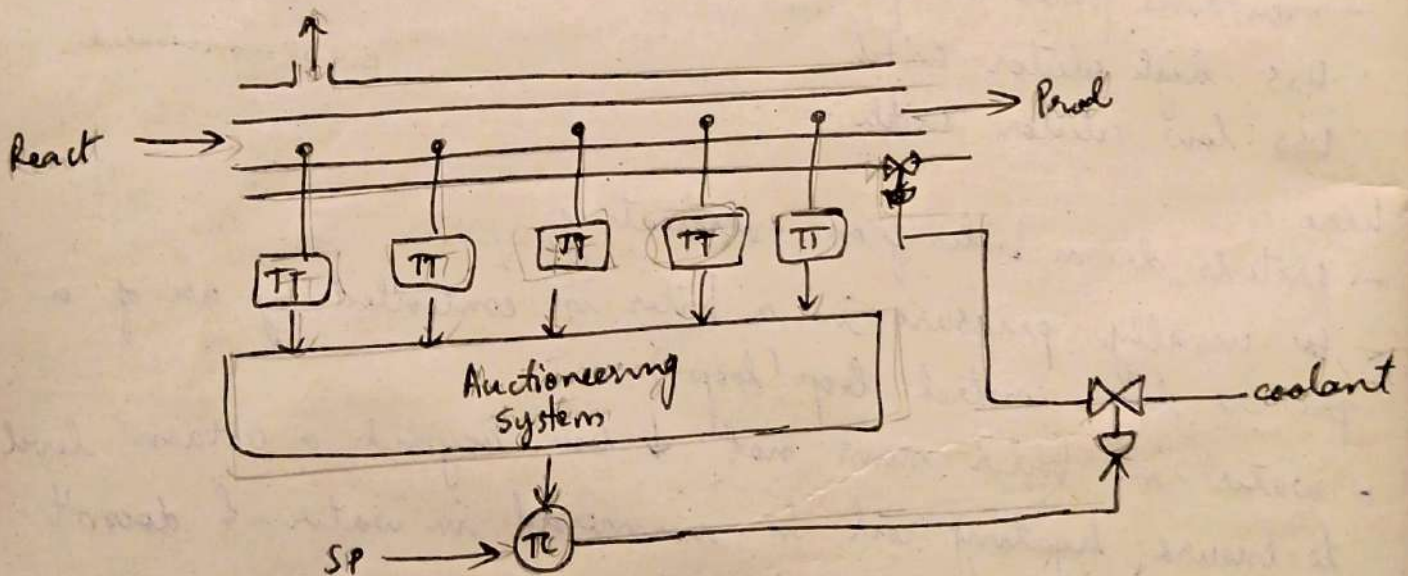
AUCTIONEERING CONTROL

- selects highest value among several similar measurements
- ie processes several measured ops & changes ~~manipulated~~ as manipulated input

Catalytic Tubular Reactors w/ Highly Exothermic Reactions



- * place thermocouples along length of reactor
- * auctioneering system to select highest temp to control coolant flow rate



In catalytic reactors, highly exothermic reactions occur. (tubular reactors filled w/ a catalytic bed) - Hot spots are the points of highest temp. They move along length of reactor as feed conditions change/catalyst activity, temp & flow of coolant, etc.

- we need to keep hot spot temp below threshold.
- CS needs to identify location of hotspot and provide proper control action

SPLIT-RANGE CONTROL

- only one measurement (controlled variable).
 - a single controlled variable is split into several parts, each affecting one of the many manipulated variables.
- eg: control of acid and base streams to control pH of a wastewater treatment

